

Original Manuscript ID: TCOM-TPS-25-2414

Article Title: “Model-Based Beam-Steered Optical Wireless Positioning with Single-LED Single-Photodiode for 3D Localization”

To: IEEE Transactions on Communications

Dear Dr. Mohammed Elamassie,

Thank you for your decision on our manuscript and for forwarding the reviewers’ constructive comments. We have carefully revised our paper according to all comments received. All modified parts in the revised manuscript are [shown in blue](#).

We would like to express our sincere gratitude to the three reviewers and to you for the thorough and insightful evaluations. The received comments have been instrumental in substantially strengthening the manuscript, both in the clarity of the exposition and in the depth of the technical analysis. In particular, they motivated us to develop a dedicated Direction Error Bound, formalize the receiver-orientation independence property, and provide a comprehensive SNR-dependent performance characterization that bridges theoretical bounds and practical estimator performance. The revised manuscript is, in our view, significantly improved as a direct result of this constructive review process.

You will find more details on the modified parts in the “Response to the Comments” document, which starts with a summary of the main modifications made to the initial paper, in response to the editor’s comments, and then addresses each reviewer’s comment individually.

Please note that all reference, figure, equation, and section numbers in the following responses refer to the revised manuscript.

Thank you very much. Looking forward to hearing from you.

Kevin Acuna-Condori and co-authors